Multimedia Signal processing

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1. **Question (8 pts)**:
   Consider a continuous source X, whose p.d.f. is given by

$$p_X(x) = e^{-x}, \quad x \geq 0 \tag{1}$$

   (a) Design a quantizer based on compander-expander. Use 1 bits to represent each symbol. Show the equivalent non-uniform quantizer in the original signal domain.

   (b) Compute the entropy of the output symbols of the quantizer.

2. **Question (5 pts)** Discuss the utility, in terms of coding efficiency, of colorspace transformations in lossless and lossy image coding.

3. **Question (5 pts)**

   (a) Illustrate the block diagram of a hybrid DCT/DPCM video codec.

   (b) Detail the pseudocode describing the sequence of processing steps applied at the encoder.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**

   (a) Let X and Y denote two continuous sources with memory, and $Q_X(\cdot)$, $Q_Y(\cdot)$ the Lloyd max quantizers designed based on the corresponding p.d.f.'s. Let $\hat{X}$ and $\hat{Y}$ the sources describing the output of the quantizers.

   - [T] [F] $\hat{X}$ and $\hat{Y}$ are discrete memoryless sources.
   - [T] [F] If X and Y have the same variance, but different p.d.f.'s (e.g., one is Gaussian and the other Laplacian), $Q_X(\cdot) = Q_X(\cdot)$
   - [T] [F] $H(\hat{X}) \leq H(X)$.

   (b) Consider an audio coding system

   - [T] [F] The analysis filterbank uses the same filters as the perceptual model.
   - [T] [F] The processing steps performed at the encoder are specified by the selected standard.
   - [T] [F] MDCT does not achieve perfect reconstruction.
   - [T] [F] The impact of frequency masking decreases with the distance from the frequency of the masker.

   (c) Consider a transform coding system

   - [T] [F] The coding efficiency always achieves that of vector quantization.
   - [T] [F] Under optimal bit allocation, more bits are assigned to coefficients having larger variance.
   - [T] [F] The DCT always maximizes the transform coding gain.

5. **Question (Matlab - 10pts)**

   Load the files *f1.mat* and *f2.mat*, which contain, respectively, the first and the second frame of the *Foreman* sequence.

   (a) Compute the entropy of the two frames.

   (b) Which coding scheme between the following is the best choice for lossless coding the second frame?

   - Intra-frame coding, in which each pixel is predicted using the pixel located on top of it
   - Inter-frame coding, in which each pixel is predicted using the same pixel in frame 1